BANGALORE INSTITUTE OF TECHNOLOGY

Department of Electronics Engineering (VDT)

PROJECT REPORT

on

ASIC IMPLEMENTATION OF A 32-BIT 8-POINT RADIX-2 FFT WITH SPI INTERFACE

(RTL to GDSII)

Design Flow: RTL Design | Functional Simulation | Logic Synthesis | Floorplanning | Placement | Routing | GDSII

Submitted by	CHARAN N
Department / Institution	Electronics Engineering (VDT), Bangalore Institute of Technology
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Tools: Synopsys VCS / DVE • Design Compiler / Design Vision • IC Compiler II

Project Overview

Table 1 Project identification

Attribute	Description
RTL top module	fft8_spi
Transform	8-point forward radix-2 FFT
Sample representation	16-bit real and 16-bit imaginary, signed Q1.15
Implementation technology	SAED32 nm RVT
Synthesis clock target	20 ns / 50 MHz
Tools	Synopsys VCS, Design Compiler and IC Compiler II
Report date	6 November 2026

This report explains the architecture, functional verification, mapped synthesis results and physical implementation of the FFT processor. The implementation record reaches a saved CTS checkpoint. Detailed routing, final physical signoff and GDSII export remain to be completed and verified.

The RTL-to-GDSII flow is presented in sequence, with measured results separated from intended downstream steps. All figures and tables are numbered for reference.

Abstract

An eight-point Fast Fourier Transform processor was implemented in Verilog with serial input and serial output through a 32-bit Serial Peripheral Interface. Each transmitted word packs a signed 16-bit real component and a signed 16-bit imaginary component in Q1.15 format. The design uses a radix-2 decimation-in-time algorithm, bit-reversed input storage and natural-order output readback. A shared complex butterfly executes four operations per stage across three stages, requiring twelve system-clock cycles for arithmetic computation. Division by two at each stage produces an overall transform scale factor of one eighth.

A self-checking VCS simulation exercised a real impulse, an imaginary constant and a delayed real impulse. All 24 output words satisfied the testbench comparison tolerance and the simulation log reported a pass. Design Compiler mapped the RTL to the SAED32 RVT library at the slow 0.95 V, 125 degree Celsius corner. At a 20 ns clock period, the mapped design contains 3,943 cells and 345 sequential cells, with a total cell area of 11,415.64 library area units. The synthesis reports show a critical setup slack of +13.24 ns, a minimum displayed hold slack of +0.16 ns and no reported transition or capacitance violations. These results cover the constrained paths of the analyzed synthesis corner.

Physical implementation in IC Compiler II includes technology and netlist import, floorplanning, power-network construction, placement and a saved clock-tree-synthesis checkpoint. Placement legality reports zero violations. At the recorded CTS checkpoint, three maximum-capacitance violations remain. The flow log reports unsupported routing options and stops before the scripted detailed-routing and export stages; the supplied GDS and SPEF directories contain no exported deliverables. Therefore, the achieved result is a verified RTL and mapped ASIC implementation with documented physical-design progress, rather than a completed tapeout-ready GDSII database.

The report analyzes fixed-point behavior, SPI synchronization, verification coverage, synthesis power and area, timing assumptions, implementation warnings and the remaining closure sequence. It concludes with practical actions for interface constraints, corner coverage, routing, electrical repair and independent physical signoff.

Keywords: ASIC, eight-point FFT, radix-2, fixed-point arithmetic, SPI, RTL, synthesis, CTS, physical design, GDSII.

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Abbreviations and notation

Table 2 Abbreviations

Term	Meaning
ASIC	Application-specific integrated circuit
FFT / DFT	Fast Fourier transform / discrete Fourier transform
DIT	Decimation in time
SPI	Serial Peripheral Interface
MOSI / MISO	Master out slave in / master in slave out
RTL / FSM	Register transfer level / finite state machine
Q1.15 / LSB	Signed fixed point with 15 fractional bits / least significant bit
CDC	Clock-domain crossing
STA / CTS	Static timing analysis / clock tree synthesis
PPA / QoR	Power, performance and area / quality of results
WNS / TNS	Worst negative slack / total negative slack
DRC / LVS	Design rule check / layout versus schematic
PG / RVT	Power and ground / regular threshold voltage
NDM / SPEF	New Data Model / Standard Parasitic Exchange Format
GDSII	Graphic Data System II layout stream format

Notation used in the analysis

x[n] denotes a complex time-domain sample and X[k] the unscaled forward DFT. Y[k] denotes the implemented scaled result X[k]/8, subject to fixed-point truncation and saturation. W8 denotes exp(-j2pi/8). Signed integer codes are converted to real values by division by 32768. Clock and timing values are in nanoseconds unless explicitly stated otherwise.

A pass or zero-violation statement applies only to the check, stage and corner named alongside it. A script command describes an intended action; a generated report or successful log entry establishes that the action occurred.

1 Introduction and project objectives

1.1 Background and motivation

The Fourier transform separates a sampled signal into frequency components. A hardware FFT reduces the arithmetic workload relative to direct DFT evaluation and provides predictable processing latency. An eight-point design is small enough to study the complete digital implementation flow while retaining the essential issues of complex multiplication, sample ordering, fixed-point scaling, timing constraints and physical interconnect.

A serial interface reduces the number of data pins needed to connect the processor to an external controller. In this implementation, the SPI host transfers eight complex input words, waits for computation and then reads eight result words. The arithmetic datapath is internal to the ASIC; the 32-bit interface width describes the packed complex transaction rather than a 32-bit precision for each component.

1.2 Objectives

The objectives are to implement a synthesizable forward FFT, define a repeatable SPI frame protocol, verify representative complex-valued responses, synthesize the design into SAED32 RVT standard cells, and carry the resulting netlist through physical implementation. The report also evaluates which results establish functional or timing correctness and which additional checks are needed before final layout release.

1.3 Scope and achieved implementation stage

The evidence includes the actual RTL and testbench, VCS simulation log, corrected synthesis reports, exported SDC, ICC2 script and session log, saved NDM checkpoints and fifteen screenshots. The implementation reaches a saved 3_cts block and a checkpoint named fft8_spi_before_cap_fix. A later detailed-routing checkpoint and exported GDSII stream are not present. Accordingly, the final stages are documented as the completion plan, not as measured achievements.

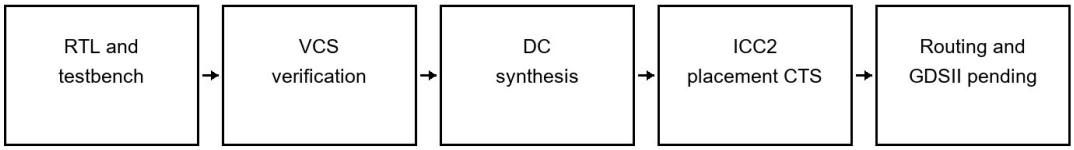


Figure 1 Implementation sequence and verified completion boundary

Figure 1 places the completed front-end stages and the recorded physical-design checkpoint in the intended RTL-to-GDSII sequence. Routing and export follow only after timing coverage and implementation-script issues have been resolved.

2 FFT theory and fixed point representation

2.1 Forward transform

The eight-point forward DFT is defined by the sum below. Frequency bin k represents the sampled frequency k times the sample rate divided by eight; bins above the Nyquist index represent negative frequencies under the usual complex-signal interpretation.

X[k] = sum(n=0..7) x[n] exp(-j 2 pi k n / 8), k = 0..7
Y[k] = X[k] / 8 before fixed-point quantization effects

For N = 8, a radix-2 decomposition has log2(8) = 3 stages and N/2 = 4 butterflies per stage. Twelve butterflies replace the direct evaluation of 64 complex products and 56 complex additions. Some FFT twiddle factors are unity or pure rotations and therefore do not require a general multiplication.

2.2 Butterfly and scaling

t = b W8^q
a_next = (a + t) / 2
b_next = (a - t) / 2

Each stage divides the two butterfly outputs by two. After three stages, the result is X[k]/8. The arithmetic right shift discards low-order bits, so the implementation introduces quantization error and can exhibit a one-code asymmetry between positive and negative outputs. Saturation limits each stored component to the signed 16-bit range.

Table 3 Numerical representation

Quantity	Representation or value
Packed sample	word[31:16] = real; word[15:0] = imaginary
Component range	-1.0 to 1.0 - 2^-15
Quantization step	2^-15 = 0.000030517578125
0.5 / 0.25 code	16384 / 8192
sqrt(1/2) coefficient	23170 / 32768, approximately 0.7070923
Overall nominal scale	1 / 8

Scaling reduces growth, but arbitrary complex inputs with both components close to full scale may still require saturation after a rotation. The supplied tests use moderate amplitudes and do not establish full-scale overflow performance.

3 Architecture and top level interface

The implemented RTL contains one top module, fft8_spi. SPI reception, synchronizers, FFT storage, arithmetic and the controller are defined within this module. The project is a frame-based serial-input/serial-output implementation, not a continuous streaming FFT pipeline. It does not contain separately instantiated u_fft and u_spi blocks or explicit RTL clock gating.

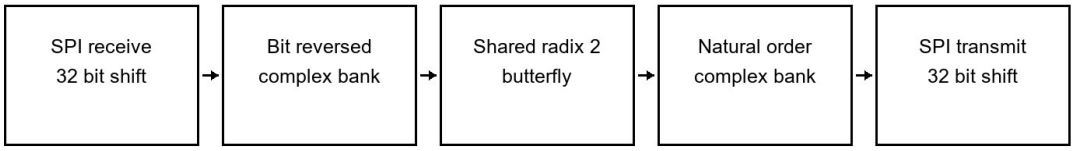


Figure 2 Functional organization derived from the implemented RTL

Table 4 Top level ports

Port	Direction	Function
clk	Input	System clock for SPI sampling and FFT arithmetic
rst_n	Input	Active-low asynchronous reset
spi_sclk	Input	External SPI serial clock
spi_cs_n	Input	Active-low select; one word per assertion
spi_mosi	Input	Serial input, most-significant bit first
spi_miso	Output	Serial output; driven low while deselected
busy	Output	High during CALC state
result_ready	Output	High during READ state

3.1 Storage and datapath reuse

The real and imaginary arrays contain eight signed 16-bit registers each, giving 256 bits of FFT sample storage. Input and output shift registers hold 32 bits each. The stage and butterfly counters select one pair of array entries per system clock, reusing the same arithmetic across all twelve butterfly operations. This reuse limits arithmetic replication at the cost of multi-cycle frame latency.

Combinational array selection and write decoding create multiplexers and control logic in the synthesized circuit. These structures contribute to the 66 logic levels reported on the synthesis critical path; the register count alone does not describe the entire implementation cost.

4 FFT ordering and arithmetic implementation

4.1 Bit reversal and stage addressing

For radix-2 DIT, natural-order incoming samples are stored at bit-reversed addresses. The three-bit index abc is converted to cba. Thus input indices 0 through 7 map to storage addresses 0, 4, 2, 6, 1, 5, 3 and 7. Once the three stages finish, the register bank contains natural-order frequency bins for direct serial readback.

Table 5 Butterfly schedule

Stage	Address pairs	Twiddle exponents
0	(0,1), (2,3), (4,5), (6,7)	0, 0, 0, 0
1	(0,2), (1,3), (4,6), (5,7)	0, 2, 0, 2
2	(0,4), (1,5), (2,6), (3,7)	0, 1, 2, 3

4.2 Complex multiplication

Re(b W) = br wr - bi wi
Im(b W) = br wi + bi wr

Four signed 16-by-16 products feed widened additions and subtractions. The multiplication results have thirty fractional bits; shifting right by fifteen restores the Q1.15 scale before the butterfly addition. The RTL widens intermediate rotation and sum values before dividing and saturating them back to sixteen bits.

Twiddle exponent zero bypasses multiplication. Exponent two applies the exact -j rotation, producing (imaginary, -real). Exponents one and three use the quantized positive or negative 23170 coefficient. These cases reduce arithmetic work while preserving the forward-transform sign convention.

4.3 Controller behavior

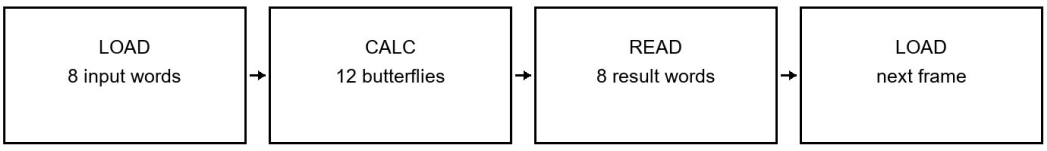


Figure 3 Frame controller sequence LOAD CALC and READ

LOAD accepts eight complete words. CALC performs one butterfly per system clock, then READ exposes eight result words. The final read returns the controller to LOAD. Input words presented during CALC are ignored; a host must respect busy and result_ready rather than assume simultaneous frame loading and readback.

5 SPI protocol and clock relationships

5.1 Transaction sequence

The slave uses SPI Mode 0: SCLK idles low, MOSI is sampled on a rising serial-clock edge and MISO advances on a falling edge. The 32-bit word is transferred most-significant bit first. Each CS assertion starts a new word, clears the receiver count and loads the outgoing shift register when a result is available. Eight write transactions form the input frame. The host then waits for result_ready and performs eight read transactions while transmitting dummy zeros.

The interface is not a shared-bus tri-state MISO implementation: the RTL explicitly drives zero while CS is high. A shared SPI bus would require an appropriate top-level output-enable or tri-state arrangement. Partial transactions and recovery from interrupted frames are not covered by the supplied directed tests.

5.2 Oversampling and synchronization

SCLK and CS pass through three-stage sampled chains and MOSI through a two-stage chain in the system clock domain. Edge pulses are derived from the synchronized SCLK history. This avoids using the external SPI clock as a second functional clock inside the datapath, but introduces sampling latency. Stable CS and MOSI timing relative to the delayed edge detection remains essential; separate synchronizers do not by themselves prove protocol-level CDC correctness.

Table 6 Clock and transfer assumptions

Item	Value or requirement
Testbench system clock	10 ns period / 100 MHz
Testbench SPI SCLK	200 ns period / 5 MHz
Synthesis system clock	20 ns period / 50 MHz
Recommended operating ratio	System clock at least 10 times SPI SCLK
CS guard intervals	At least four system cycles before first edge and between words
Arithmetic at 50 MHz	12 cycles, nominal 240 ns, excluding interface latency

The 100 MHz functional simulation does not demonstrate 100 MHz physical timing closure; synthesis was constrained at 50 MHz. At 5 MHz SPI, sixteen 32-bit transactions require at least 102.4 microseconds of clocked bit time per transform frame, excluding guard intervals and computation. This gives an idealized upper bound below about 9.77 thousand transforms per second. The serial link dominates frame throughput.

6 Functional verification methodology

The testbench tb_fft8_spi instantiates the DUT, produces a 100 MHz clock, applies active-low reset and drives Mode 0 SPI transactions. The spi_transfer task asserts CS, shifts a complete word, samples MISO and inserts guard delays. The check_bin task interprets the two output halves as signed integers and permits an absolute error of one code in each component. An independent timeout terminates a stalled simulation.

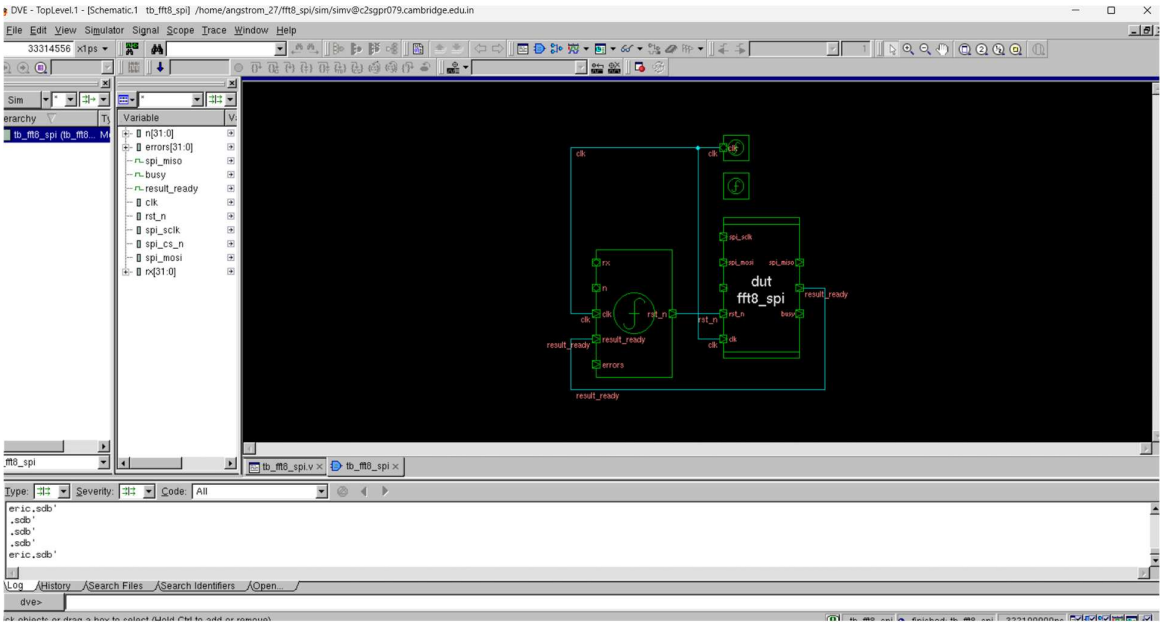


Figure 4 DVE testbench hierarchy and DUT interface

Table 7 Directed verification cases

Case	Stimulus	Expected scaled response
Real impulse	x[0] = 0.5; all others zero	All eight bins = 0.0625 + j0
Imaginary DC	Every sample = j0.25	Bin 0 = j0.25; other bins zero
Delayed real impulse	x[1] = 0.5; all others zero	Magnitude 0.0625 with bin-dependent phase

The impulse checks gain and distribution, the imaginary constant checks component handling and DC cancellation, and the delayed impulse checks the forward sign convention, nontrivial twiddles and bin ordering. These three cases provide focused diagnostic coverage; they are not exhaustive verification of all inputs or SPI recovery conditions.

7 Simulation waveform evidence

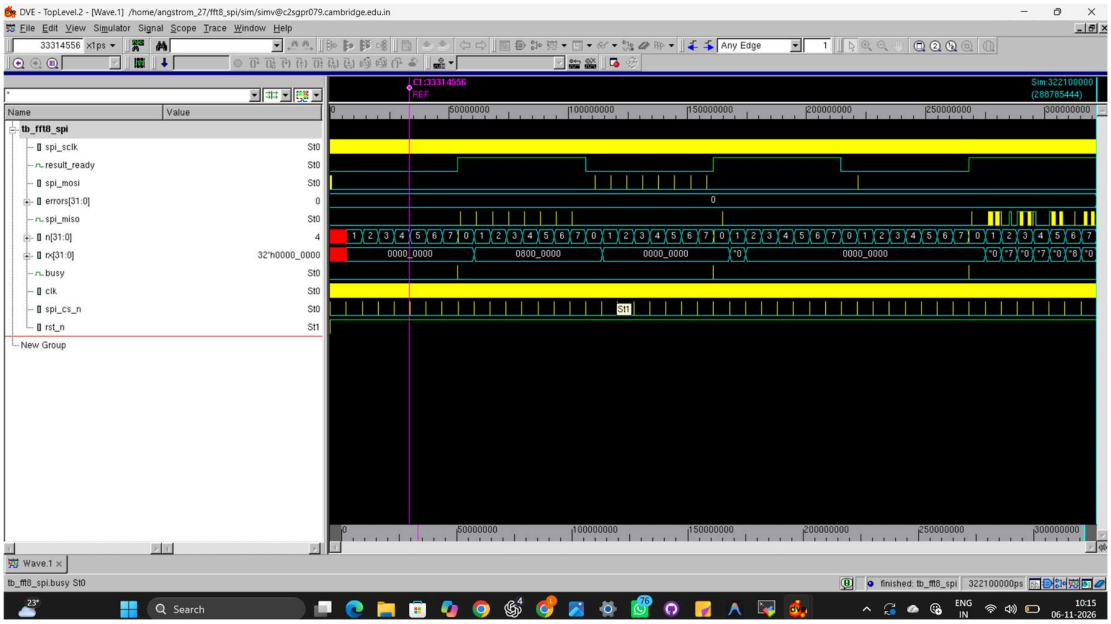


Figure 5 Complete DVE waveform capture for the directed simulation

The waveform shows repeated load and read intervals, result_ready windows and short busy pulses for arithmetic calculation. The errors variable remains zero in the capture. The serial clocks appear as dense bands at the displayed time scale; individual Mode 0 setup and hold behavior would require a closer time-axis view.

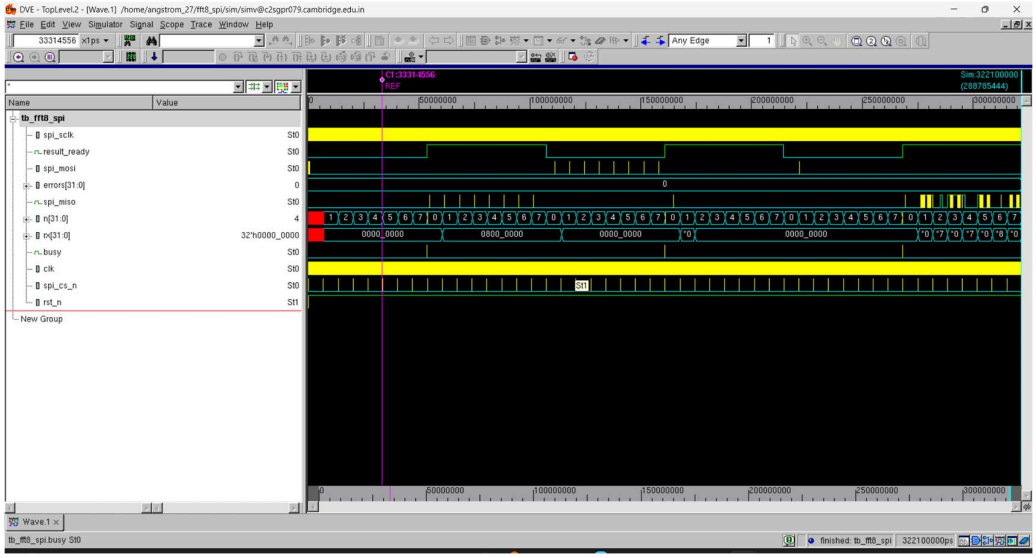


Figure 6 Alternate capture of the same simulation interval

The two captures show the same verification run and are retained together for traceability. They are not counted as independent tests. The simulation log, rather than a visual estimate from these waveforms, establishes the exact values of the returned bins.

8 Functional results and numerical analysis

The VCS simulation log reports all three tests passed and ends at 322,100,000 ps, equivalent to 322.1 microseconds. Twenty-four complex result words were read and checked. For the first impulse, every result is 0x08000000. For imaginary DC, bin zero is 0x00002000 and the other seven bins are zero.

Table 8 Delayed impulse output from simulation

Bin	Real code	Imaginary code	Packed hexadecimal
0	2048	0	08000000
1	1448	-1449	05A8FA57
2	0	-2048	0000F800
3	-1449	-1449	FA57FA57
4	-2048	0	F8000000
5	-1448	1448	FA5805A8
6	0	2048	00000800
7	1448	1448	05A805A8

For bin one, the ideal components are approximately +1448.1547 and -1448.1547 in Q1.15 integer units. The observed 1448 and -1449 values are consistent with coefficient quantization and arithmetic right-shift truncation. Bin five returns -1448 rather than the testbench nominal -1449 and remains within the permitted one-code tolerance. This distinction matters when reporting exact numerical agreement.

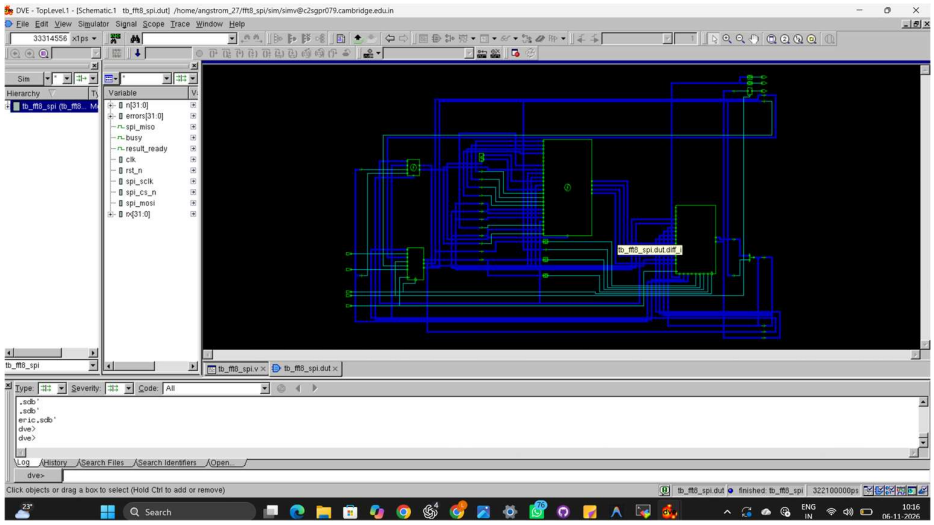


Figure 7 Elaborated RTL datapath connectivity shown in DVE

This elaborated schematic exposes arithmetic and signal connectivity. It is evidence of the RTL structure, not of standard-cell physical placement or post-route equivalence.

9 Logic synthesis setup and constraints

Design Compiler T-2022.03-SP5 reads rtl/fft8_spi.v, elaborates fft8_spi and maps the logic with compile_ultra. The corrected target library is /opt/PDK/DBs/saed32rvt_ss0p95v125c.db. The original -40 degree Celsius filename was unavailable and caused an earlier generic, unmapped run; only the corrected mapped results are used in the performance tables.

Table 9 Synthesis configuration

Parameter	Setting
Target library	saed32rvt_ss0p95v125c
Process and voltage	Slow corner / 0.95 V
Temperature	125 degrees Celsius
Clock period and waveform	20 ns; rising at 0 ns and falling at 10 ns
Setup uncertainty	0.1 ns
Hold uncertainty	0.01 ns
Clock latency	0.3 ns
Clock transition	0.1 ns
Maximum capacitance constraint	200 fF at design level
Reset timing exception	False path from rst_n

9.1 Interpretation of timing margins

The period sets the functional frequency target at 50 MHz. Setup uncertainty reserves time before the next capture edge, while hold uncertainty increases the minimum-delay requirement around the capture edge. Latency represents an assumed clock arrival offset; it is not the period. Before CTS, a common ideal latency on both launch and capture can cancel in a register-to-register calculation.

9.2 Exported handoff files

The corrected run generates output/fft8_spi_HDL.v, output/fft8_spi.ddc and const/fft8_spi.sdc. These provide the mapped netlist, saved synthesis database and timing constraints for implementation. A successful file write alone does not establish timing completeness; the SDC must still match the external interface and intended operating modes.

10 Logical hierarchy and schematic review

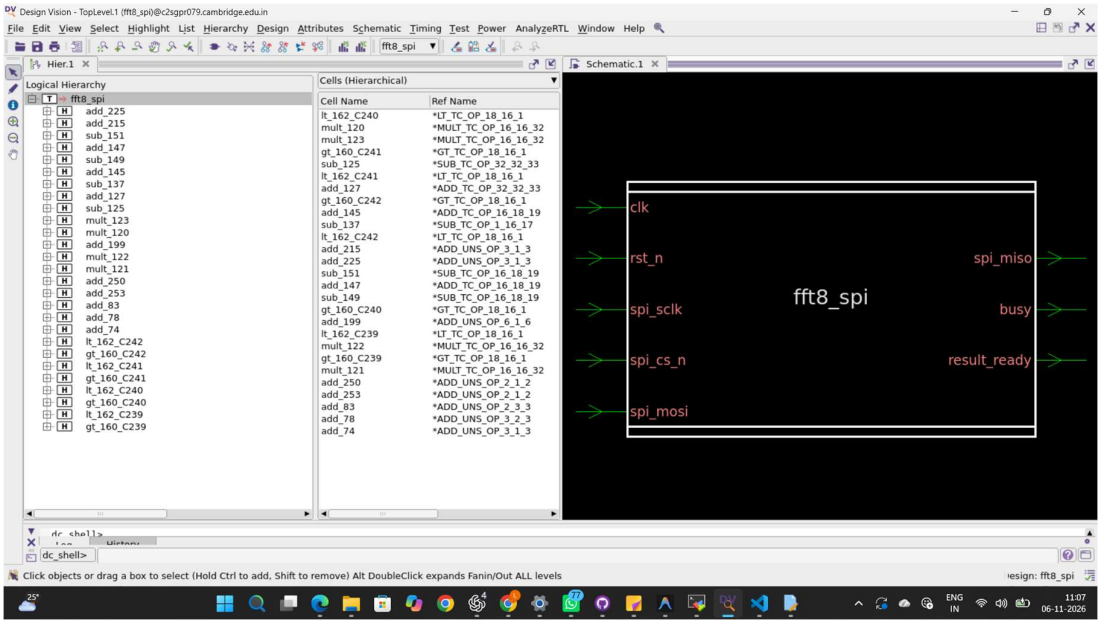


Figure 8 Design Vision top symbol and logical operator hierarchy

The Design Vision capture identifies fft8_spi and its eight ports. The hierarchy pane contains adders, subtractors, comparators and multipliers derived from RTL elaboration. Operator names in a GUI view do not establish that the displayed view is the final flattened mapped netlist. The area report confirms actual SAED32 cell mapping for the corrected synthesis run.

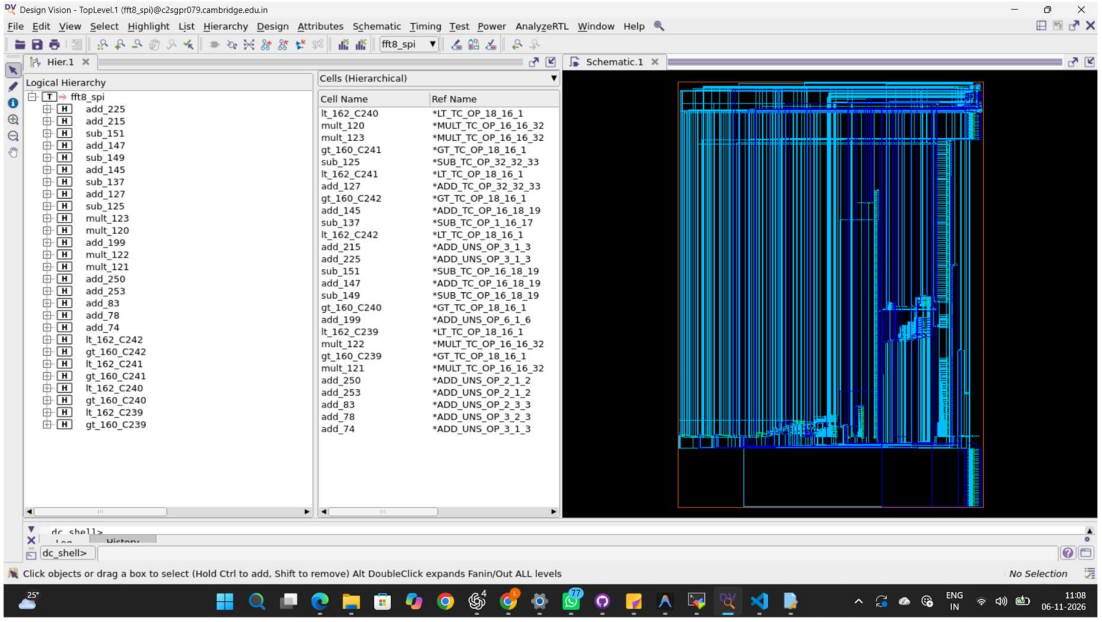


Figure 9 Expanded Design Vision connectivity view

The expanded schematic illustrates the large amount of internal connectivity created by sample selection, arithmetic and control. Cell-level placement and routing must resolve these connections physically. The screenshot does not provide a quantitative congestion or path-delay measurement.

11 Synthesis area and power results

Table 10 Mapped synthesis cell and area results

Metric	Reported value
Total cells	3,943
Combinational cells	3,598
Sequential cells	345
Buffers and inverters	719, included in combinational count
Buffer / inverter breakdown	57 / 662
Nets / top-level ports	4,526 / 8
Combinational area	8,960.609091
Noncombinational area	2,455.031083
Total cell area	11,415.640173
Estimated net area	3,116.822732
Total estimated design area	14,532.462905

Area values are retained in the library report units. The cell area is the sum of mapped cell areas, whereas the estimated design area includes a wire-load-based net-area estimate. Neither value is the final die area or measured core utilization. Buffers and inverters are subsets of the combinational count and must not be added again to the total cell count.

Table 11 Synthesis power estimate

Component	Power
Cell internal	98.6211 microW
Net switching	1.0617 microW
Total dynamic	99.6828 microW
Cell leakage	509.7292 microW
Total estimate	609.4117 microW, approximately 0.6094 mW

Leakage accounts for approximately 83.6 percent of the reported total. The power report warns that primary inputs and sequential outputs are unannotated, so activity is estimated rather than derived from a representative workload. This is a synthesis estimate at the specified corner, not a measured chip power result. Post-CTS and post-route power may differ due to clock buffering, interconnect and actual switching activity.

12 Synthesis timing and quality of results

Table 12 Timing results at the synthesis corner

Metric	Result
Clock period	20 ns
Critical path length	6.63 ns
Critical setup slack	+13.24 ns
Setup total negative slack	0 ns
Setup violating paths	0
Minimum displayed hold slack	+0.16 ns
Hold violating paths	0
Maximum transition violations	0
Maximum capacitance violations	0
Critical path logic levels	66

12.1 Setup interpretation

A positive setup slack means that the analyzed path arrives before the required capture time under the applied model. The +13.24 ns result provides considerable margin against the 20 ns target. It must not be converted directly into a claimed maximum operating frequency: the result includes specific constraints, a single cell corner and wire-load estimates rather than extracted routed parasitics.

12.2 Hold interpretation

The displayed minimum path starts and ends at bit_count_reg[0] through an AO22 cell. Its arrival time is 0.48 ns and its required time is 0.32 ns, producing +0.16 ns slack. This demonstrates a met hold check at the synthesis corner. Fast-corner timing and propagated clock effects remain separate checks for physical implementation.

12.3 Mapping correction and report validity

The earlier failed library configuration produced zero cell delay, zero area and artificial -0.01 ns hold results in generic cells. Those reports are superseded by the corrected SAED32 mapping. Keeping report sets separate prevents invalid generic results from being mistaken for ASIC timing data.

The corrected synthesis is a sound starting point for physical design, but its zero-violation result applies to paths that are actually constrained. The following section explains the remaining coverage gaps.

13 Timing coverage and verification limitations

13.1 External interface coverage

The exported SDC contains no input-delay or output-delay commands. The synthesis timing check identifies spi_sclk, spi_cs_n and spi_mosi as missing input timing, and busy, result_ready and spi_miso as unconstrained output endpoints. The first stages of the sampled SPI inputs are also unconstrained. A zero-violation timing report does not establish correctness for these paths.

Because SPI is oversampled, external serial inputs should be handled with an explicit asynchronous-interface timing and CDC strategy. Any false-path exception should be narrowly scoped to the intended first synchronizer stage and reviewed against the protocol. The direct CS-to-MISO masking path and the transmit timing relative to the external host require their own external timing budget. Applying arbitrary synchronous input delays merely to suppress warnings would not establish correctness.

13.2 Reset and synchronizer implementation

The RTL uses asynchronous reset on its state and storage registers. The SDC false-path exception removes reset paths from normal timing analysis. This does not establish safe reset deassertion; reset-domain-crossing behavior, recovery/removal assumptions and reset distribution must be checked at integration. Synchronizer recognition and preservation should also be reviewed in implementation so that CDC chains receive appropriate placement treatment.

13.3 Checks generated before synthesis

The script writes check_design and check_timing before compile_ultra. Eight unused-cell warnings in check_design therefore describe the pre-optimization design. They may be optimized away, but a fresh post-compile check is needed to establish the final status. The stale warnings should not be reported as either unresolved final defects or automatically fixed issues without rerunning the check.

13.4 Scope of functional confidence

The three directed tests check representative behavior with a one-LSB tolerance. Additional tests should cover arbitrary complex inputs, saturation, negative extremes, CS interruption, reset during transfer, back-to-back frames and operation near the minimum clock ratio. Formal equivalence between RTL and mapped netlist is not established by the available simulation log.

14 Physical design environment and floorplanning

IC Compiler II U-2022.12-SP3 imports the synthesized netlist and links it to saed32_rvt.ndm. The import log confirms 3,943 instances, 4,526 nets and eight top-level ports. The intended physical flow uses an NDM design library named fft8_spi.ndm, a SAED32 technology file and TLU+ early and late parasitic models.

Table 13 Physical implementation setup

Item	Recorded configuration
Design library	fft8_spi.ndm
Reference library	/opt/PDK/CLIBs/saed32_rvt.ndm
Technology	/opt/PDK/tech/saed32nm_1p9m.tf
Core utilization target	0.7, or 70 percent
Core offset argument	{10 10}
Pin placement	place_pins -self
Mode / corner / scenario	func / slow / func_slow
Late / early parasitic models	MAX_TLU / MIN_TLU

14.1 Floorplan interpretation

The floorplan defines the rows, core area and boundary space required for cells, pin access and PG structures. The 70 percent value is a requested initialization utilization, not a measured final occupancy. Optimization, CTS and filler insertion can change the instance population. Final utilization must be measured from the implemented block rather than inferred from the script alone.

14.2 Technology and layer warnings

The ICC2 log reports missing preferred routing directions and automatically derives alternating horizontal and vertical directions for routing layers. Earlier technology checks also report pitch-related warnings. These messages require review against the lab PDK configuration; successful import does not certify the physical technology setup. Layer mapping, extraction and reference-cell compatibility must remain consistent through stream-out.

No standalone floorplan screenshot is available. The later implementation figures show the core boundary and its PG context; they are labeled according to their actual saved checkpoint rather than being reassigned to an earlier floorplanning stage.

15 Power planning and physical organization

Table 14 Power network geometry requested in the script

Structure	Layer assignment	Geometry
Core ring	Horizontal M7; vertical M8	Width 2; spacing 3; offset {3 3}
Mesh	Vertical M6; horizontal M5	Width 0.5; pitch 10; interleaved nets
Standard-cell rails	M1	VDD and VSS connections
Power domain	Default domain	Single-domain VDD and VSS

The ring provides a perimeter distribution path, the mesh carries supply across the core and the M1 rails connect the cell rows. Geometry values are script settings in technology-defined distance units. They are not results of an IR-drop or electromigration optimization. Connectivity, via coverage and electrical robustness require separate checks.

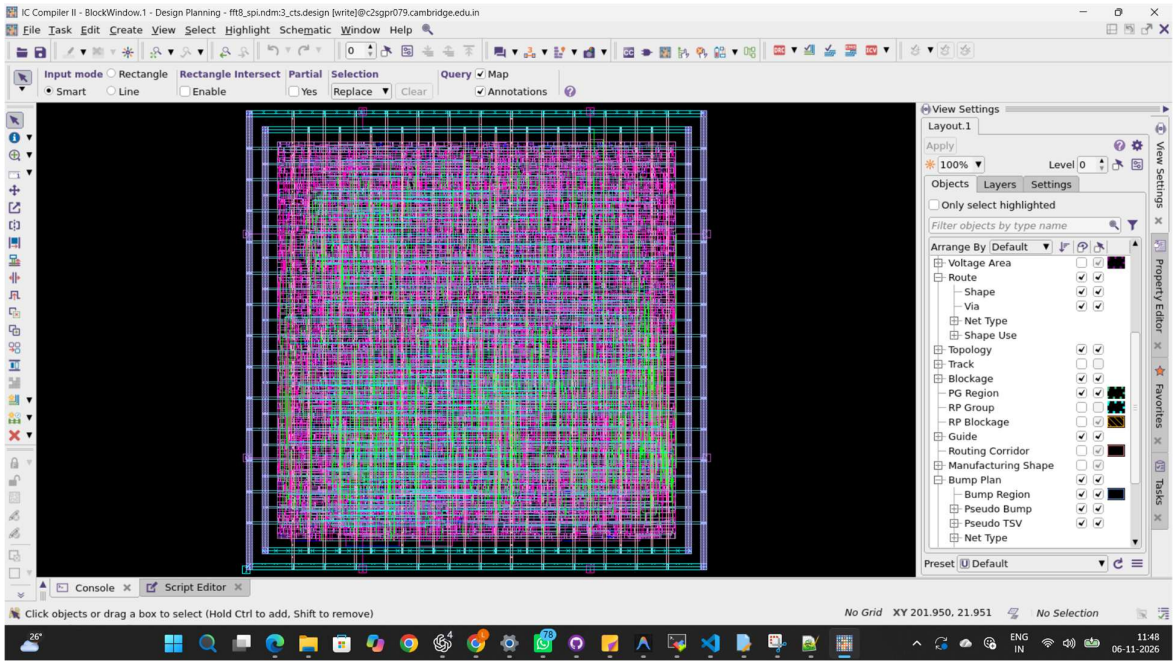


Figure 10 Saved 3_cts checkpoint showing core and interconnect layers

The screenshot title explicitly identifies the 3_cts saved block. Dense geometry includes power, clock and other implementation layers; the image is not evidence that the later scripted route_auto stage completed. The script stops before that command in the available session log.

15.1 Verification required for the power network

The flow requests PG DRC, connectivity and missing-via checks. Their final standalone reports are absent from the supplied output set, so complete PG closure is not claimed. The visible VDD/VSS domain provides structural context but cannot replace connectivity reports or an extracted supply integrity analysis.

16 Placement legality and congestion

The implementation sequence requests initial placement, legalization and place_opt. The placement_legality report states that check_legality succeeded and records zero violations across overlap, site alignment, orientation, spacing, region and other checked placement categories. This supports a legal placed design at the time the check was run.

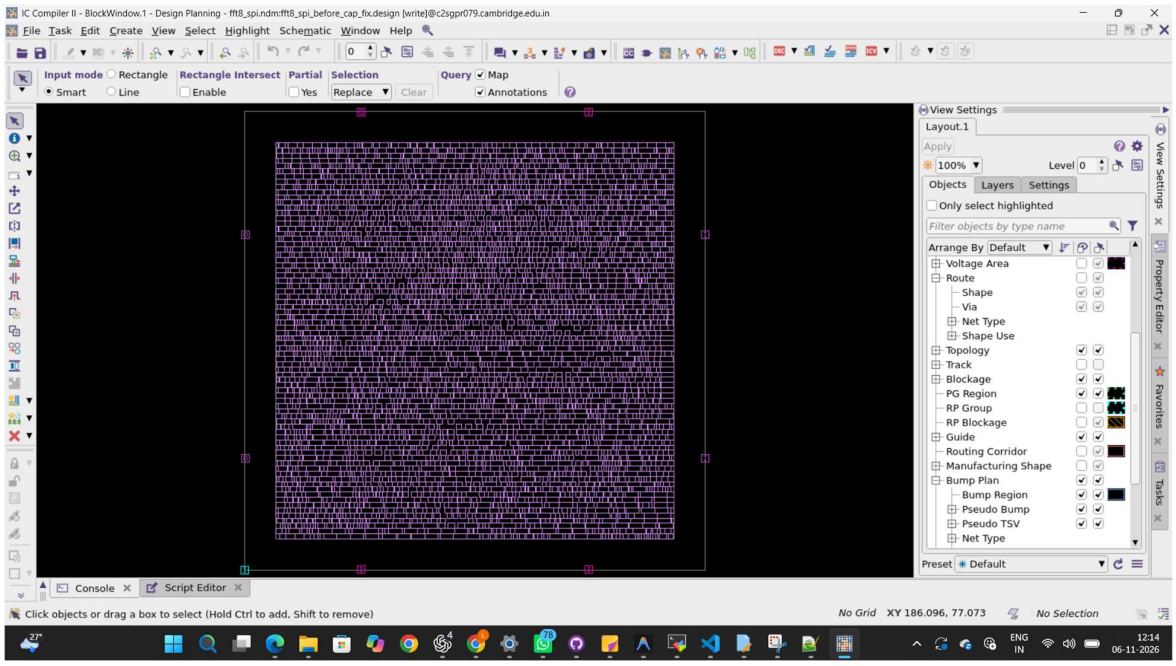


Figure 11 Cell placement view of the before capacitance repair checkpoint

Although captured later in the session, this figure isolates the standard-cell arrangement for placement discussion. Its title is fft8_spi_before_cap_fix, not 2_placed. The visible rows and cell outlines are therefore a later checkpoint viewed with selected layers hidden.

Table 15 Placement congestion report snapshot

Direction	Total overflow	Maximum	Overflowing GRCs
Both	169	8	113 / 0.73 percent
Horizontal	90	6	78 / 1.01 percent
Vertical	79	8	35 / 0.45 percent

The congestion report explicitly warns that the map is out of date and recommends rerunning the global router. These values are a historical placement snapshot; they do not establish final routing congestion or DRC closure. The legality report is also a placement check and must not be substituted for final foundry DRC.

17 Clock tree synthesis checkpoint

17.1 Recorded CTS outcome

The ICC2 session runs clock_opt and saves 3_cts. The final clock-opt QoR summary records zero negative setup and hold paths, zero maximum-transition violations and three maximum-capacitance violations. It lists an area of 11,704.86 and 3,880 instances, including 276 buffers and 638 inverters. These figures are implementation-stage results and should not replace the separate synthesis cell and area totals.

Table 16 Recorded clock opt summary

Metric	Value
Negative setup paths / total negative slack	0 / 0
Negative hold paths / total hold violation	0 / 0
Maximum transition violations	0
Maximum capacitance violations	3
Area / instance count	11,704.86 / 3,880
Buffers / inverters	276 / 638
Saved stage	3_cts

17.2 Completion warning and timing coverage

The log also reports FLW-8004, Incomplete clock_opt / final_opto. A saved CTS checkpoint therefore does not prove that all clock optimization phases finished cleanly. WNS equal to zero in the violation summary indicates no negative slack listed; it is not a measurement of available positive timing margin. Actual clock latency, skew and full maximum/minimum paths must be reported for the intended scenarios.

Repeated warnings say that no clock routing rules were specified. The flow should define or explicitly review the intended clock routing layers and rules. It should also confirm that post-CTS timing uses propagated clock behavior and that any pre-CTS latency assumptions are handled consistently. The 0.1 ns setup and 0.01 ns hold uncertainties are maintained, but their adequacy depends on the integration clock budget.

17.3 Single corner limitation

The script creates one scenario named func_slow, but a scenario name alone does not demonstrate full timing-library corner binding. Early and late TLU+ models characterize interconnect; they do not substitute for fast and slow standard-cell timing views. Required multi-corner setup and hold analysis remains part of the completion work.

18 Implementation layout evidence

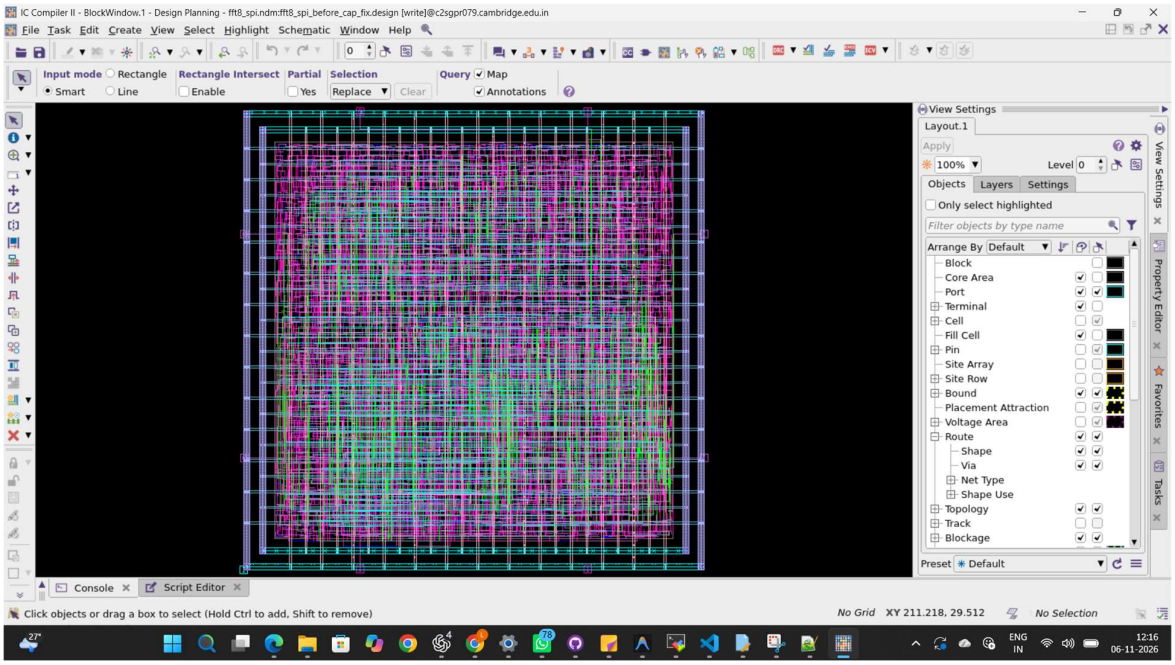


Figure 12 Before capacitance repair checkpoint with power and interconnect overlay

This view adds layer geometry to the cell arrangement shown in Figure 11. The outer supply structures and the internal network are visible. Since the flow log stops before route_auto, the screenshot is described as an implementation checkpoint rather than a final routed layout.

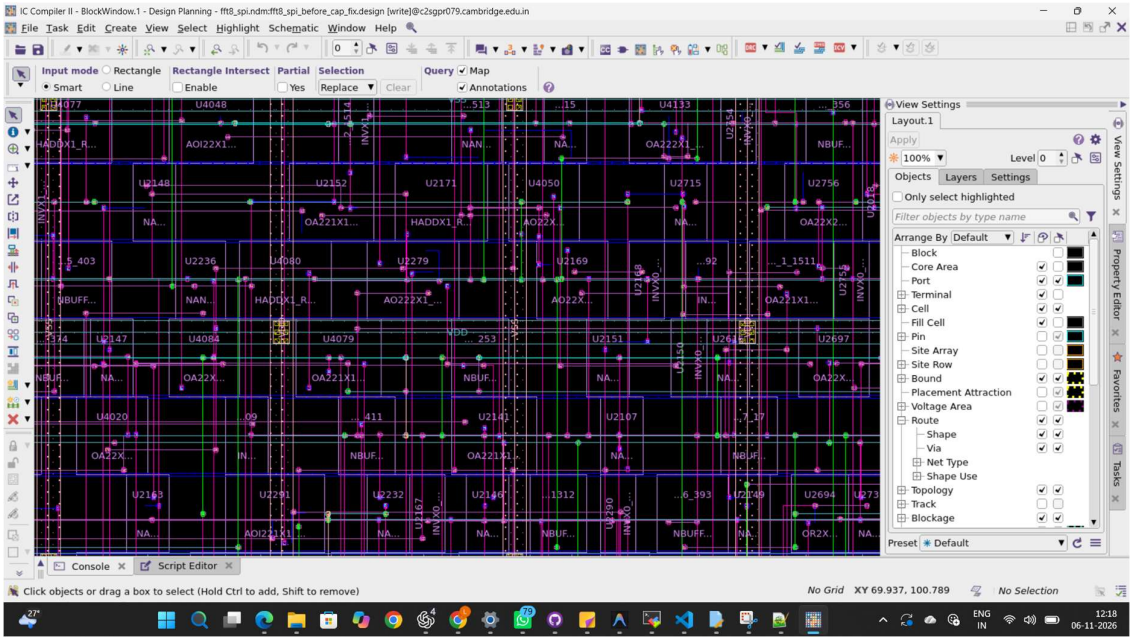


Figure 13 Detailed view of standard cells pins and local interconnect

The close-up shows named cells, pin shapes, vias and interconnect around individual rows. Such a view helps debug local pin access and connectivity, but cannot quantify opens, shorts, antenna effects or spacing violations without the corresponding route and signoff reports.

19 Layout overview and schematic correspondence

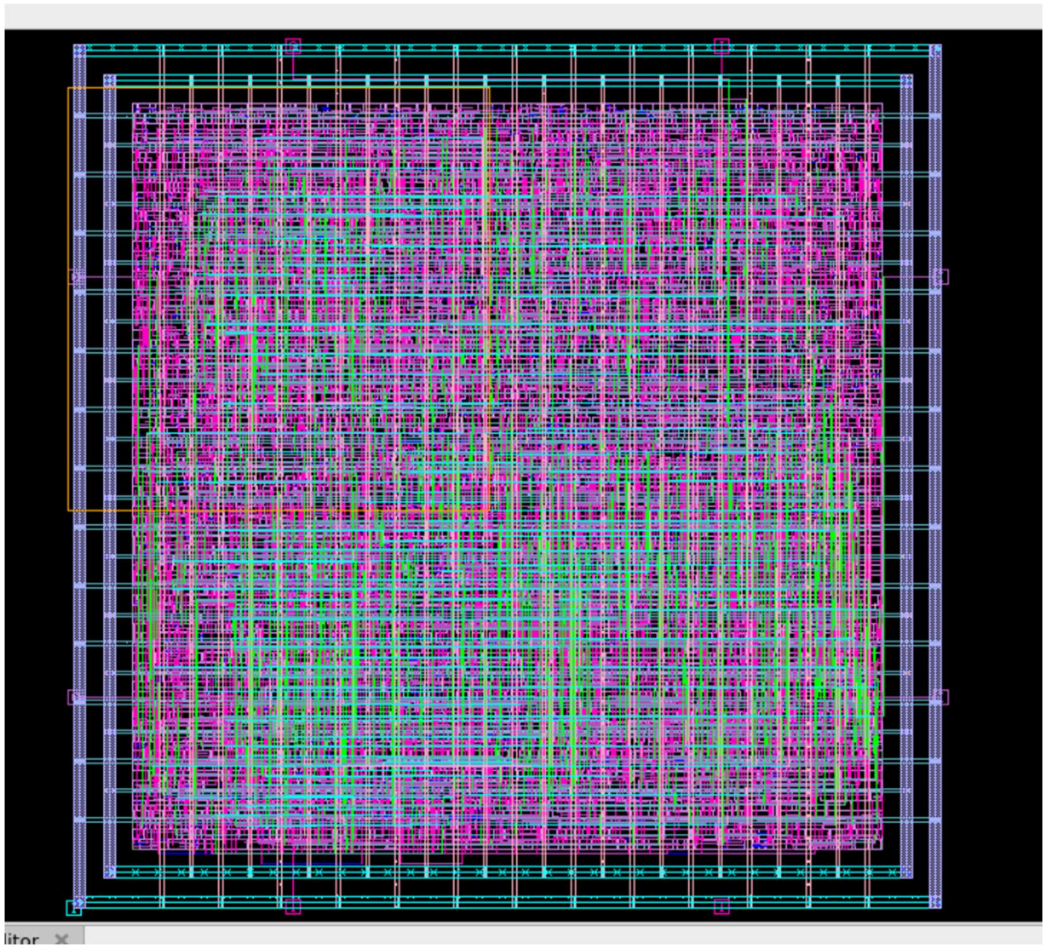


Figure 14 Enlarged overview of the implemented layout checkpoint

The cropped overview provides a clearer view of the core and surrounding PG geometry than the full desktop captures. It does not include a checkpoint title, so its stage is interpreted together with the adjacent timestamped ICC2 captures and session log. No additional routing completion is inferred from the density of displayed lines.

The core appears regular and approximately rectangular, with cells organized in rows and multi-layer structures spanning the active region. Final die size, routed wire length and density cannot be extracted accurately from this image and are not assigned numerical values in this report.

20 Physical and logical correspondence

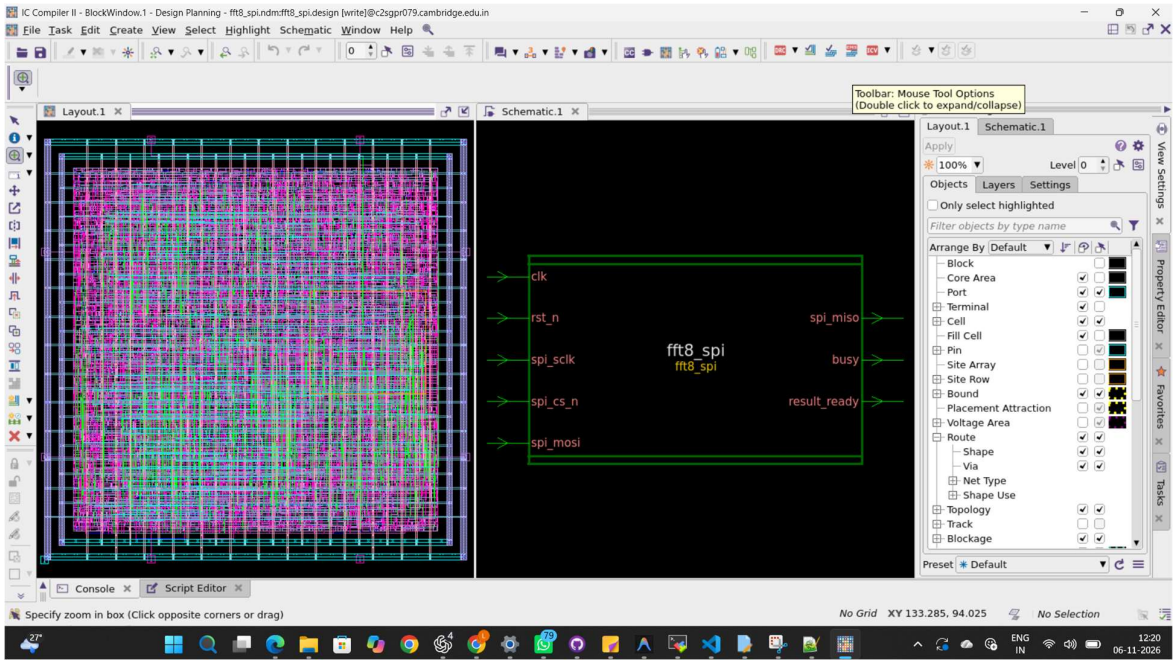


Figure 15 ICC2 layout alongside the fft8_spi top level symbol

The side-by-side view associates the physical block with the same eight-signal top-level interface used in RTL and synthesis. It is useful for confirming design identity and navigation between physical and logical representations. It is not an RTL-to-layout equivalence proof.

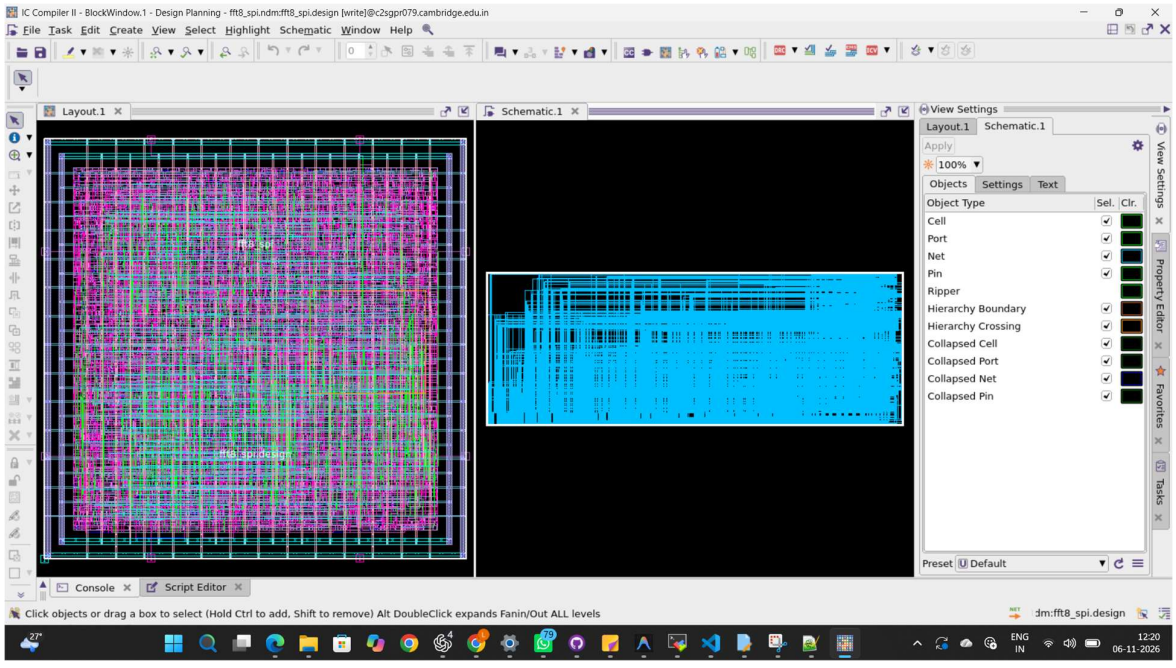


Figure 16 ICC2 layout and expanded connectivity in the same design session

The expanded connectivity view shows the complexity of the implemented logic relative to its compact interface. It does not establish timing or connectivity closure; those properties are evaluated using reports and independent verification rather than visual appearance.

21 Power domain and implementation checkpoint

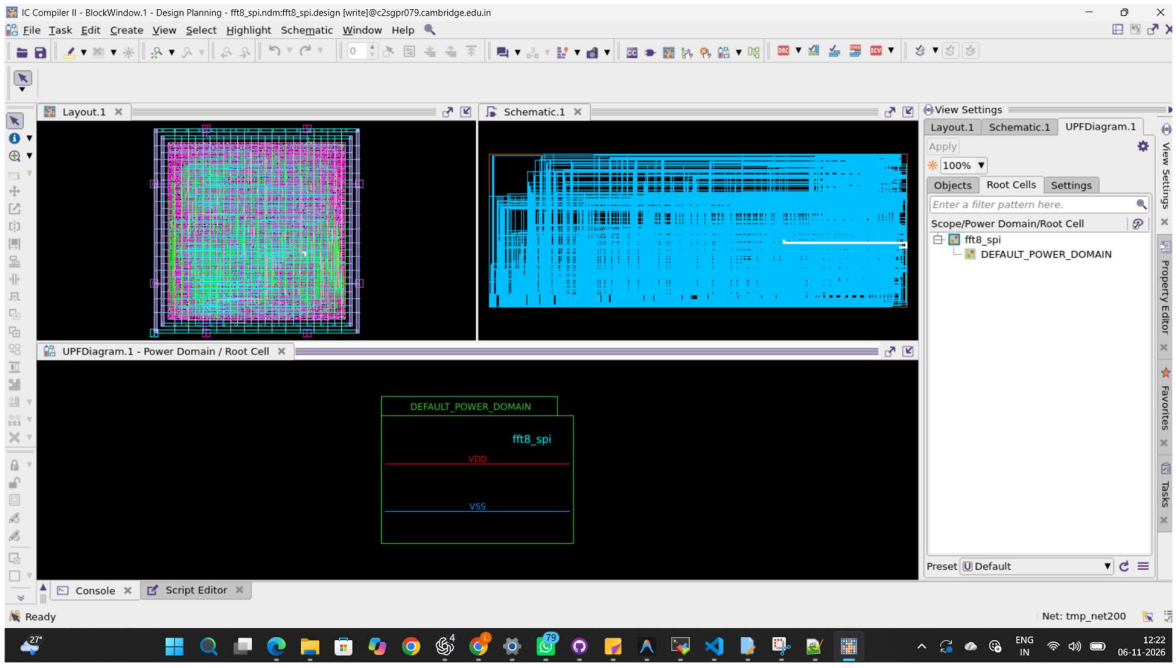


Figure 17 ICC2 layout schematic and default power domain views

The power-domain panel identifies fft8_spi in DEFAULT_POWER_DOMAIN with VDD and VSS. This supports a single-supply organization in the displayed database. It does not show a multi-voltage power intent implementation, isolation, retention or power gating, and no such features are claimed for this project.

21.1 Saved block evidence

Table 17 Saved NDM blocks found in the project

Block	Interpretation
1_init	Initialization handoff checkpoint
2_placed	Placed design checkpoint
3_cts	Saved checkpoint after clock_opt
fft8_spi	Working design block
fft8_spi_before_cap_fix	Checkpoint saved before electrical repair

The checkpoint names establish what was saved, not that every associated implementation task passed. The directory set does not contain 4_routed, fft8_spi_after_cap_fix or fft8_spi_final. Existing blocks should be reopened rather than recreated with the initialization script, which deletes fft8_spi.ndm at startup.

22 Routing interruption and remaining violations

22.1 Recorded interruption

After saving 3_cts, the script reports CMD-104 for route.common.connect_within_datum and route.detail.repair_shorts because those names are not application variables in the installed release. It then rejects route.detail.drc_convergence_mode with CMD-013 and stops p1.tcl at line 200. The later route_auto, route_opt, filler insertion, final reports and export commands are not demonstrated as executed.

Error: Invalid option name 'route.detail.drc_convergence_mode'
script .../pd1/p1.tcl stopped at line 200 due to error

```
Number of max_transition violation(s): 0

Mode: func Corner: slow
Scenario: func_slow
max_capacitance
                          Required        Actual
Net                      Capacitance    Capacitance       Slack  Violation
-------------------------------------------------------------------------
n2895                        8.00           8.01          -0.01  (VIOLATED)
  PIN : U3576/Y              8.00           8.01          -0.01  (VIOLATED)

n3420                        8.00           8.00          -0.00  (VIOLATED)
  PIN : U2123/Y              8.00           8.00          -0.00  (VIOLATED)

n1294                       16.00          16.00          -0.00  (VIOLATED)
  PIN : ctmTdsLR_1_1414/Y
                            16.00          16.00          -0.00  (VIOLATED)

-------------------------------------------------------------------------
Number of max_capacitance violation(s): 3


Mode: func Corner: slow
Scenario: func_slow
-------------------------------------------------------------------------
Number of min_capacitance violation(s): 0

Mode: func
Corner: slow
Scenario: func_slow

-------------------------------------------------------------------------
Number of min_pulse_width violation(s): 0
Total number of violation(s): 3
```

Figure 18 ICC2 constraint report showing three maximum capacitance violations

The displayed report lists zero transition, minimum-capacitance and minimum-pulse-width violations, but three maximum-capacitance violations. These are electrical loading violations, not a count of physical routing DRC errors. Small negative slack remains a violation even when it rounds to -0.00 at the displayed precision.

23 Capacitance analysis and corrective actions

Table 18 Remaining capacitance violations

Net	Driving pin	Limit	Displayed load	Slack
n2895	U3576/Y	8.00	8.01	-0.01
n3420	U2123/Y	8.00	8.00	-0.00
n1294	ctmTdsLR_1_1414/Y	16.00	16.00	-0.00

The three loads exceed the applicable maximum-capacitance limits. Two values appear equal to their limits because the report rounds the underlying numbers. The clock-related instance name on n1294 suggests a CTS-created element, but its function must be confirmed from the net and cell attributes rather than inferred from the name alone. Capacitance units should be confirmed in ICC2 before comparing this table to fF values from synthesis.

23.1 Repair strategy

Start from the saved checkpoint and inspect each driver, fanout and physical wire loading. Candidate implementation repairs include selecting a stronger allowed driver, buffering or splitting the load, or reducing excessive route length. Clock-network nets require clock-aware repair so that skew and latency are not degraded. These are proposed remedies, not changes already executed in the project.

First remove or replace unsupported options using the installed ICC2 command help and review the incomplete clock_opt warning. Resume the appropriate optimization stage, then run supported routing and electrical optimization commands. Do not change the library capacitance limits or relax uncertainty merely to eliminate reported violations.

23.2 Recheck after repair

Rerun constraint reporting with more numerical precision, full setup and hold timing, route connectivity and physical checks. Record the active mode, cell corners and parasitic models. Save a new block after verified repair, preserving the original checkpoint for comparison. A new checkpoint name alone does not establish closure; attach the reports generated from that exact block.

24 GDSII handoff and signoff requirements

24.1 Present export status

The PD script requests routed Verilog, SPEF and GDSII generation, but the supplied gds and spef folders contain no files. The available session log stops before those commands and no final routed checkpoint is present. GDSII stream-out is therefore pending. The project title describes the intended flow; it must not be read as a claim that the implementation is tapeout ready.

Table 19 Intended final handoff artifacts

Artifact	Purpose	Current evidence
fft8_spi_routed.v	Implemented netlist	Not present in export folder
fft8_spi.spef	Extracted parasitics	Not present
fft8_spi.gds	Layout stream	Not present
Final NDM block	Reopenable implemented database	Final named block not present
Route and signoff reports	Connectivity, timing and geometry checks	Final report set not present

24.2 Required completion sequence

Complete scenario and interface timing setup, resolve clock_opt completion and unsupported commands, repair electrical violations, execute detailed routing and optimization, and verify final connectivity and timing with extracted parasitics. Insert the appropriate physical-only cells and apply the PDK-required checks. Then export using the validated GDS layer map and matching reference-cell geometry.

A final release should include setup and hold analysis across the required corners, final DRC and LVS using the appropriate verification decks, antenna checks, PG connectivity and via coverage, and relevant reliability analysis. An ICC2 check_lvs invocation or an empty all-violators timing table does not replace independent foundry-rule physical signoff.

24.3 Export reproducibility

Record the exact saved block, tool release, reference libraries, extraction models and layer map used for export. Reimport or inspect the generated stream to confirm top-cell identity and hierarchy resolution. Archive the netlist, SDC, parasitics, layout stream and reports as one matched release rather than combining artifacts from different checkpoints.

25 Consolidated results and engineering discussion

Table 20 Measured results and completion boundaries

Stage or metric	Result	Interpretation
RTL tests	3 frames; 24 output words; pass	One-LSB component tolerance
Synthesis mapping	SAED32 RVT successful	Corrected SS 0.95 V 125 C library
Synthesis clock	50 MHz	20 ns constraint
Synthesis setup / hold	+13.24 ns / +0.16 ns	Constrained single-corner paths
Synthesis cells / cell area	3,943 / 11,415.64	Area in library units
Synthesis power	0.6094 mW estimated	Activity not workload annotated
Placement legality	0 violations	Checked placement stage
CTS electrical limits	3 max-cap violations	Still unresolved
Detailed routing / GDSII	Not verified / no export found	Script interruption before routing

25.1 Throughput and resource tradeoff

The shared butterfly reduces arithmetic duplication and makes the twelve-cycle processing cost small compared with serial transfer time. At the 50 MHz synthesis target, the nominal arithmetic interval is 240 ns, whereas the SPI data phase alone takes 102.4 microseconds at 5 MHz for one complete write-read frame. Improvements to internal compute latency would have little impact on sustained host throughput unless the serial protocol or buffering also changes.

25.2 Interpretation of the implementation results

Mapped synthesis establishes a realizable standard-cell representation with substantial setup margin under the selected model. Placement legality confirms a valid row-based arrangement. The physical log demonstrates meaningful implementation progress, but electrical and timing coverage issues remain. The layout screenshots provide useful structural evidence and cannot establish detailed-route, DRC, LVS or GDS export completion on their own.

26 Recommendations and future scope

26.1 Immediate implementation work

Complete interface constraints from the actual host timing specification. Document the system-to-SPI clock ratio, CS guard periods, readback latency and output loading. Review synchronizer placement, asynchronous reset release and the direct CS masking path. Rerun post-synthesis and post-CTS timing checks so that each remaining unconstrained path has an explicit, justified treatment.

Review the installed ICC2 option names rather than retaining options from another flow or release. Correct the clock_opt completion issue, add the required timing corners and verify cell-corner selection independently of early/late RC model selection. Repair the three capacitance violations through implementation changes and regenerate constraint reports before downstream export.

26.2 Verification improvements

Add a bit-accurate reference model and randomized complex samples with explicit scale and saturation behavior. Exercise full-scale extremes, negative limits, arbitrary reset timing, interrupted CS, malformed word lengths and the minimum clock-ratio boundary. Check no data are lost or duplicated across repeated frames and confirm host behavior when data are presented while busy.

Run equivalence between RTL and mapped netlist, and check the final implementation against the synthesized functional representation. Representative switching activity from realistic input frames would improve dynamic-power estimates. Spectral error, quantization bias and saturation rate should be measured for workloads beyond the three diagnostic vectors.

26.3 Architectural extensions

Double buffering could overlap input loading with result readback or computation if the protocol is extended. A wider or faster interface would improve throughput more directly than reducing an already short butterfly schedule. Parameterized transform sizes, selectable scaling and rounding-to-nearest could broaden use, but require fresh verification and physical characterization.

Explicit clock gating should be considered only after activity analysis establishes a benefit and a supported gating flow is verified. The present design does not implement the clock-gating architecture described in the reference report, so no such savings are attributed to this project.

27 Source records and reproducibility

Project results are traceable to the following records. File paths in this section are relative to the supplied fft8_spi project directory unless reports2 is explicitly named. The earlier reference PDF informs report organization; its numerical results and architecture are not substituted for this implementation.

Table 21 Primary evidence records

Reference	Record	Use in this report
[R1]	rtl/fft8_spi.v	Actual FFT, SPI and controller architecture
[R2]	tb/tb_fft8_spi.v	Directed stimuli, comparison tolerance and clock generation
[R3]	sim/simulation.log	24 observed words and test pass result
[R4]	reports2/synth_area.rpt and synth_qor.rpt	Mapped cell counts, area and timing summary
[R5]	reports2/synth_timing*.rpt	Setup and minimum-delay path evidence
[R6]	reports2/synth_power.rpt	Power estimate and activity warnings
[R7]	const/fft8_spi.sdc	Exported clock settings and exceptions
[R8]	pd1/p1.tcl and icc2_output.txt	Physical flow settings and interruption
[R9]	reports/placement_legality.rpt and congestion.rpt	Placement checks and dated congestion estimate
[R10]	fft8_spi.ndm block directories	Saved implementation checkpoint inventory
[R11]	15 screenshots dated 6 November 2026	RTL, synthesis and physical views
[R12]	ASIC_FFT8_SPI_RTL_to_GDSII_Report.pdf	Reference document organization only

Tool versions recorded in the logs are VCS U-2023.03, Design Compiler T-2022.03-SP5 and ICC2 U-2022.12-SP3. Source and report timestamps are used to distinguish functional simulation, corrected synthesis and the later physical-design session. No device measurements or fabricated-silicon results are claimed.

28 Conclusion

The project demonstrates a functional eight-point radix-2 FFT processor with serial input and output over a 32-bit SPI transaction format. The word packs two signed Q1.15 components. Bit-reversed input storage, shared butterfly arithmetic, exact special-case twiddle rotations and stage scaling produce natural-order outputs representing approximately one eighth of the forward DFT. The twelve-cycle arithmetic schedule provides a compact frame-oriented implementation.

The self-checking simulation passes three directed frames and twenty-four output comparisons within the declared one-code tolerance. Corrected Design Compiler synthesis maps the processor to SAED32 RVT cells at a 20 ns clock target, yielding 3,943 cells and 11,415.64 library units of cell area. Reported setup and hold slack are positive for the analyzed constrained paths, and the synthesis electrical-rule summary is clean. The total power estimate is approximately 0.6094 mW, with the limitations of unannotated switching activity.

Physical implementation successfully imports the mapped design, establishes a floorplan and power structures, produces legal placement and saves a CTS checkpoint. The available records retain three capacitance violations and an incomplete clock_opt warning. Unsupported routing options stop the scripted flow before the demonstrated execution of detailed routing and GDSII export. These findings define a precise completion boundary and avoid conflating a visually dense layout with verified final physical closure.

The next implementation steps are to finish external timing and CDC/reset review, validate scenario and corner setup, correct the ICC2 command sequence, complete clock optimization, repair loading violations and run detailed routing. Final extracted timing, connectivity, geometry and signoff checks must accompany a validated GDSII export. With those actions, the existing verified RTL and mapped implementation provide a practical foundation for completing the intended RTL-to-GDSII flow.